EDS 222 Midterm

Fall 2026

Name:

You have **75 minutes**. The exam is worth **100 points**:

- **Part 1:** five short questions, 10 points each. Aim to spend about 25 minutes on Part 1.
- **Part 2:** two analysis questions, 25 points each. Aim to spend about 20 minutes on each.

Write your answers in the space provided. If you run out of room, continue on the back of the page and say so. Show your work on calculations, because partial credit is available. Use the variable names given in each question when you write R code or model equations.

Question	Points	Score
1. Fill in the blanks	10	
2. Fix the error	10	
3. Model representations	10	
4. Interpreting inference	10	
5. Causal diagrams	10	
6. Analysis	25	
7. Analysis	25	
Total	**100**	

Reference

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right) \qquad p = \frac{1}{1+e^{-\text{logit}(p)}}$$

$$F = \frac{(\text{RSS}_{simple} - \text{RSS}_{complex}) \;/\; (p_{complex} - p_{simple})}{\text{RSS}_{complex} \;/\; (n - p_{complex})}$$

Backdoor criterion. A set of variables satisfies the backdoor criterion if adjusting for them (1) closes every backdoor path between x and y, and (2) doesn't include any variables caused by x.

Part 1: Short questions

1. Fill in the blanks (10 points)

Ecologists counted invertebrate taxa (`richness`) in 24 streams in **Unlogged** (reference level) and **Logged** watersheds (`logging`). The data are in `streams`.

(a) (6 points) Fill in the blanks to build the null distribution and calculate a two-sided p-value.

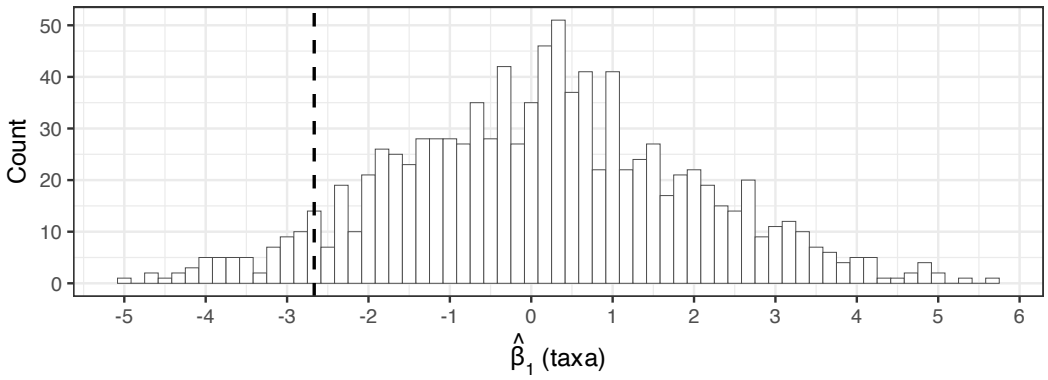
```
beta_1_hat <- coef(lm(richness ~ logging, data = streams))["loggingLogged"]

null_dist <- replicate(1000, {
  shuffled <- mutate(streams, logging = __(1)__(logging))
  shuffled_lm <- lm(richness ~ logging, data = __(2)__)
  coef(shuffled_lm)["loggingLogged"]
})

p_value <- mean(abs(null_dist) >= abs(__(3)__))
```

(1) _____ (2) _____ (3) _____

(b) (4 points) The histogram shows `null_dist`. The dashed line is the observed $\hat{\beta}_1$ = -2.67, and the p-value is 0.172. Shade the area of the histogram that corresponds to the p-value.



2. Fix the error (10 points)

Biologists recorded whether each of 150 snowy plover nests hatched (`hatched`: 1 = yes, 0 = no) and the number of beachgoers per hour nearby (`visitors`). The data are in `nests`. Each part has **exactly one** error. Describe it and *briefly* explain how to fix it. (5 points each.)

(a) Fit a logistic regression of hatching on visitors.

```
plover_glm <- glm(hatched ~ visitors, data = nests)
```

(b) The p-value for `visitors` is 0.004.

> "There is a 0.4% chance that visitors have no effect on hatching."

3. Model representations (10 points)

Researchers counted eelgrass `shoots` per m² at 40 sites, along with mean summer water `temperature` (°C) and wave `exposure` (**Sheltered**, the reference level, or **Exposed**). They fit `lm(shoots ~ temperature * exposure, data = eelgrass)`.

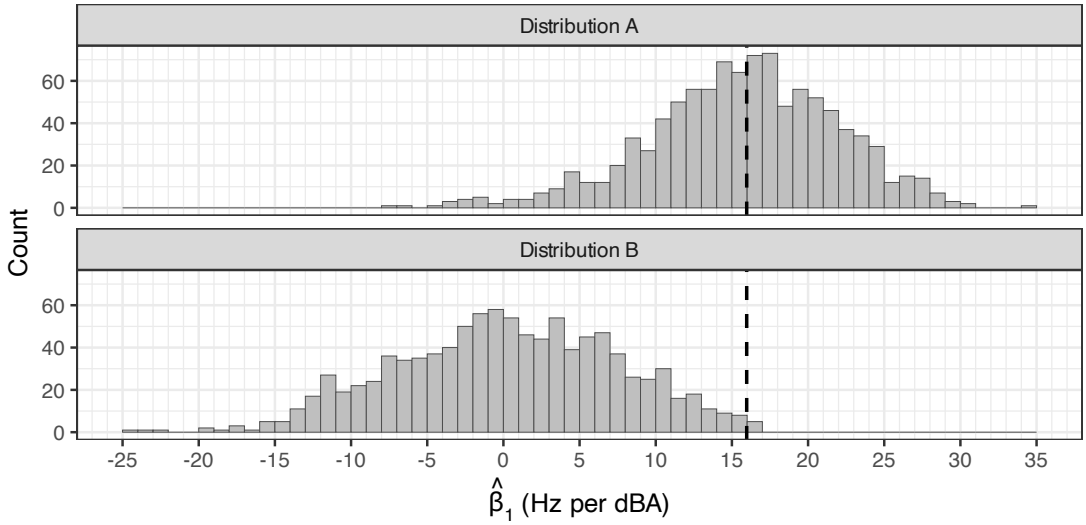
	Estimate
(Intercept)	562.58
temperature	-15.60
exposureExposed	-7.61
temperature:exposureExposed	-5.50

(a) (6 points) Write the model in statistical notation, filling in the coefficient values using the estimates above. You can write σ for the residual standard deviation.

(b) (4 points) What is the slope of `temperature` for **sheltered** sites? What is the slope of `temperature` for **exposed** sites? Show your work.

4. Interpreting inference (10 points)

Researchers fit `lm(song_freq ~ noise)` to 30 sparrow territories, relating song pitch (Hz) to traffic noise (dBA). The figure shows two simulations of $\hat{\beta}_1$: one **shuffled** `noise`, the other **resampled rows with replacement**. The dashed line is the observed $\hat{\beta}_1 = 16.0$.



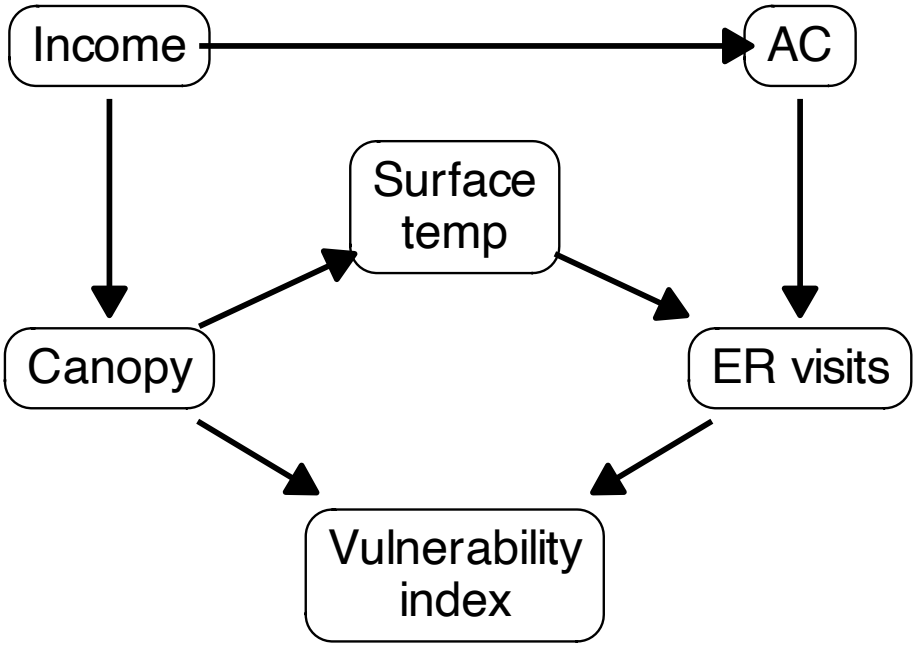
(a) (3 points) Which distribution came from shuffling? How can you tell?

(b) (4 points) Is the p-value closer to 0.1 or 0.01? Do you reject H_0 at $\alpha = 0.05$?

(c) (3 points) Use the figure to approximate the standard error of $\hat{\beta}_1$.

5. Causal diagrams (10 points)

A public health agency asks: **how does tree canopy affect heat-related emergency room (ER) visits?** Here is their DAG for 80 neighborhoods. The city calculates each neighborhood's heat "Vulnerability index" from its canopy cover and its heat-related ER visits.



(a) (6 points) Is each variable a confounder, mediator, or collider for the effect of Canopy on ER visits?

Income: _____ Surface temp: _____ Vulnerability index: _____

(b) (4 points) To estimate the **total effect** of canopy on ER visits, which variables should be in the model besides Canopy? Why?

Part 2: Analysis questions

6. Phosphorus and algae in lakes (25 points)

Phosphorus fertilizes algae, and too much algae turns lakes green, depletes oxygen, and can produce toxins. A state agency sampled 60 lakes, 20 in each of three watershed land uses (`land_use`): **Forest** (reference level), **Agriculture**, and **Urban**. They measured total **phosphorus** (µg/L) and **chlorophyll** *a*, a measure of algae (µg/L).

Agricultural and urban watersheds send more phosphorus into lakes, from fertilizer and stormwater. Land use also affects chlorophyll through other pathways: lakes in cleared watersheds have less shoreline shade, so they're warmer and sunnier, which helps algae grow. Deeper lakes (`depth`) have less chlorophyll, but depth isn't related to land use or phosphorus. Finally, the agency visits lakes more often (`monitoring`, visits per year) if they have high phosphorus or high chlorophyll.

Question: how does phosphorus affect chlorophyll?

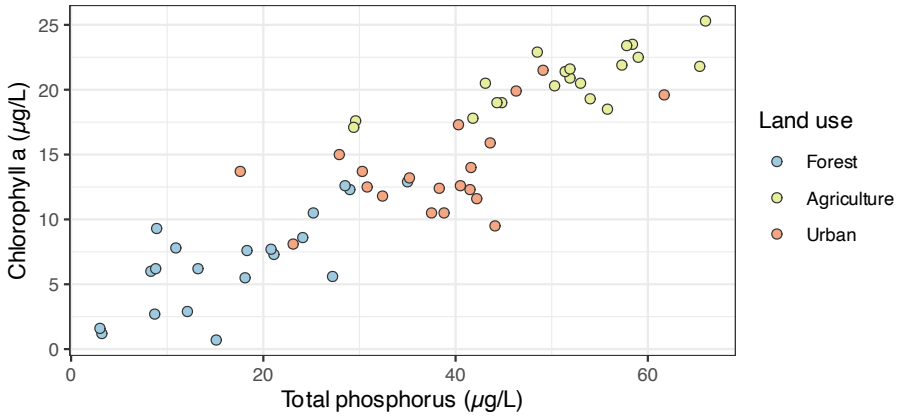
Variable	Description
`chlorophyll`	Chlorophyll *a* (µg/L)
`phosphorus`	Total phosphorus (µg/L)
`land_use`	Watershed land use: Forest (reference level), Agriculture, or Urban
`depth`	Maximum lake depth (m)
`monitoring`	Monitoring visits per year

(a) DAG (6 points). Draw a DAG with all five variables in the table.

(b) Choose variables (6 points). Which variables should you include in the model, and which should you leave out? For each variable, say why, using the terms confounder, collider, and mediator where they apply.

(c) Model (4 points). Write your model in statistical notation (use βs, not numbers), and define any dummy variables. Then write the formula you'd put in `lm()`.

Here's the data.



Three analysts each fit a different model.

```
lake_a <- lm(chlorophyll ~ phosphorus, data = lakes)
coef(summary(lake_a))
```

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)    1.460     0.8824    1.66 1.03e-01
phosphorus     0.349     0.0226   15.43 3.41e-22
```

```
lake_b <- lm(chlorophyll ~ phosphorus + land_use, data = lakes)
coef(summary(lake_b))
```

```
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)          3.010     0.7939    3.79 3.70e-04
phosphorus           0.221     0.0339    6.52 2.17e-08
land_useAgriculture  6.532     1.3800    4.73 1.54e-05
land_useUrban        2.344     1.0550    2.22 3.04e-02
```

```
lake_c <- lm(chlorophyll ~ phosphorus + land_use + monitoring, data = lakes)
coef(summary(lake_c))
```

```
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)        -0.9958     0.6360  -1.566 1.23e-01
phosphorus          0.0136     0.0297   0.459 6.48e-01
land_useAgriculture 2.6517     0.9328   2.843 6.27e-03
land_useUrban       0.6004     0.6692   0.897 3.74e-01
monitoring          1.1147     0.1144   9.741 1.42e-13
```

```
anova(lake_a, lake_b)
```

```
Analysis of Variance Table

Model 1: chlorophyll ~ phosphorus
Model 2: chlorophyll ~ phosphorus + land_use
  Res.Df RSS Df Sum of Sq    F  Pr(>F)
1     58 498
2     56 335  2       163 13.7 1.5e-05
```

(d) Inference (5 points).

i. (1 point) Which model matches your answer to (b)? Use it for the rest of this question.

ii. (2 points) State H_0 and H_A for the effect of phosphorus on chlorophyll, and interpret the p-value for `phosphorus` in a sentence.

iii. (2 points) The `anova()` output compares `lake_a` and `lake_b`. What are its H_0 and H_A? What do you conclude?

(e) Interpretation (4 points).

i. (2 points) Interpret the coefficient on `phosphorus` from your model in a sentence, with units.

ii. (2 points) Predict the chlorophyll in an urban lake with 40 µg/L of phosphorus. Show your work.

7. Drought and tree mortality (25 points)

During California's 2012–2016 drought, bark beetles killed more than 100 million trees in the Sierra Nevada. Forest ecologists revisited 300 ponderosa pines in 2017 and recorded whether each tree had died (`dead`: 1 = dead, 0 = alive). For each tree, they also had the climatic water **deficit** during the drought (hundreds of mm; higher means more drought stress), the tree's **elevation** (km), and its radial **growth** (mm/yr) in 2014, partway through the drought.

Lower elevations are hotter, so they had larger deficits. Lower elevations also have milder winters, so more bark beetles survive there to attack trees. Drought stress slowed trees' growth, and slow-growing trees are less able to defend themselves against beetles. Drought stress also killed trees directly.

Question: how did drought stress (`deficit`) affect the probability that a tree died?

Variable	Description
`dead`	1 if the tree died during the drought, 0 if it survived
`deficit`	Climatic water deficit during the drought (100s of mm)
`elevation`	Elevation (km)
`growth`	Radial growth in 2014 (mm/yr)

(a) DAG (6 points). Draw a DAG with all four variables.

(b) Choose variables (5 points). We want the **total effect** of drought stress on mortality. Which variables should you include, and which should you leave out? Say why, using the terms confounder, collider, and mediator where they apply.

(c) Model (4 points). Write your model in statistical notation (use βs, not numbers). Include the distribution of the response. Then write the formula you'd put in `glm()`, and its `family` argument.

Three analysts each fit a different model.

```
tree_a <- glm(dead ~ deficit, family = binomial(link = "logit"), data = trees)
coef(summary(tree_a))
```

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept)    -6.27      0.754   -8.32 8.93e-17
deficit         1.17      0.150    7.83 4.86e-15
```

```
tree_b <- glm(
  dead ~ deficit + elevation,
  family = binomial(link = "logit"),
  data = trees
)
coef(summary(tree_b))
```

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   -3.823      2.288   -1.67  0.09472
deficit        0.931      0.259    3.60  0.00032
elevation     -0.760      0.678   -1.12  0.26232
```

```
tree_c <- glm(
  dead ~ deficit + elevation + growth,
  family = binomial(link = "logit"),
  data = trees
)
coef(summary(tree_c))
```

```
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   -0.869      2.615  -0.332   0.7396
deficit        0.649      0.283   2.292   0.0219
elevation     -0.862      0.680  -1.268   0.2048
growth        -0.922      0.411  -2.244   0.0248
```

(d) Inference (5 points).

i. (1 point) Which model matches your answer to (b)? Use it for the rest of this question.

ii. (2 points) State H_0 and H_A for the effect of deficit on mortality, in words and in terms of a β.

iii. (2 points) Interpret the p-value for `deficit` in a sentence. Do you reject H_0?

(e) Interpretation (5 points).

i. (2 points) Interpret the coefficient on `deficit` from your model in a sentence, with units.

ii. (3 points) Calculate the probability that a tree at 1.5 km elevation with a deficit of 6 (600 mm) died. Show your work.